

Brain serotonin content: physiological regulation by plasma neutral amino acids.

When plasma tryptophan is elevated by the injection of tryptophan or insulin, or by the consumption of carbohydrates, brain tryptophan and serotonin also rise; however, when even larger elevations of plasma tryptophan are produced by the ingestion of protein-containing diets, brain tryptophan and serotonin do not change. The main determinant of brain tryptophan and serotonin concentrations does not appear to be plasma tryptophan alone, but the ratio of this amino acid to other plasma neutral amino acids (that is, tyrosine, phenylalanine, leucine, isoleucine, and valine) that compete with it for uptake into the brain.